

$(\mathbf{R}(w_1, m_1), \dots, \mathbf{R}(w_{L_s}, m_{L_s}))$. Effectively, we keep a phrase of length L_p unchanged for each sentence, and replace other tokens with random tokens. We compute $\mathcal{P}_{\text{REP-P}}(w) = \mathcal{M}(w | [\tilde{s}^1; \tilde{s}^2; \dots; \tilde{s}^{n-1}; \mathbf{R}(w_1, m_1), \dots, \mathbf{R}(w_{i-1}, m_{i-1})])$ and the average token probability that only considers the kept tokens $\tilde{\text{TP}}_N = \frac{1}{L_p} \sum_{i=L_s-L_p+1}^{L_s} \mathcal{P}_{\text{REP-P}}(w | w = w_i, n = N) \times m_i$. As a concrete example, we take a sentence $s = (\text{Apple}, \text{is}, \text{red})$ and $m = (0, 1, 1)$, and the target token $w = \text{red}$. Then, the demonstrations discussed will be like ‘Apple is red // Orange is red // Banana is red’. The pattern we kept unchanged is in color red. In the context of in-context learning, this pattern corresponds to demonstrations like ‘...Answer: D’ or ‘...Response: Positive’.

Figure 10 depicts our experimental results for repeated and discontinuous phrases. We use randomly generated sentences here. We see that disrupting the sentence-level loop with random prefixes does not break the self-reinforcement effect. The effect persists across various LLaMA models. In addition, we observe a stronger reinforcement when increasing the phrase length. More tokens are kept unchanged, higher probability the model assigns to the phrase. Particularly, Figure 10(D) serves as an ablation study, as each line progressively adds one more unchanged token. A repeating phrase of ‘...XYZ’ gets reinforced more compared to the phrase ‘...XY’, indicating that each repetitive token bolsters its subsequent tokens.

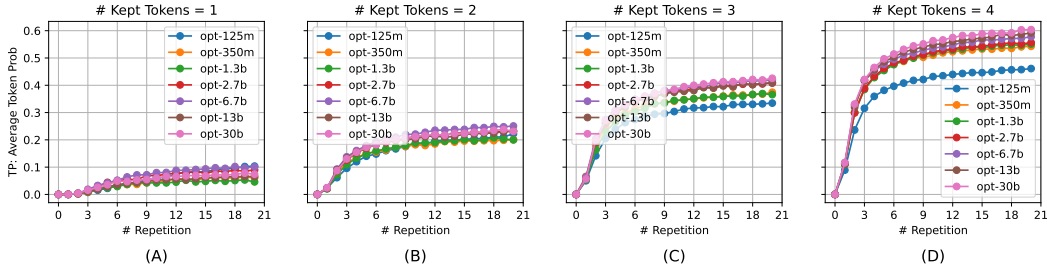


Figure 11: Token co-occurrence reinforcement of OPT models.

D.2 TOKEN REINFORCEMENT OF OPT MODELS

In Figure 11, we plot the token reinforcement of all 7 OPT models. The results are consistent with our results of LLaMA in the main context, validating the generality of token reinforcement across different families of large language models.

D.3 TOKEN REINFORCEMENT OF OTHER LLMS

We present token reinforcement on LLaMA-2 (Touvron et al., 2023b)(7B, 13B, 70B), Mistral (Jiang et al., 2023)(7B), GPT-J (Wang & Komatsuzaki, 2021)(6B). Figure 12 demonstrates that all these LLMs exhibit similar token reinforcement effect, with various strength. These results consolidate our findings.

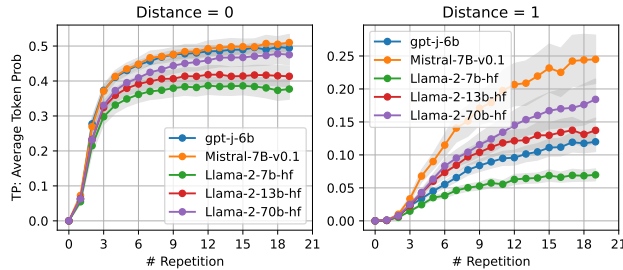


Figure 12: Token reinforcement on Various LLMs.

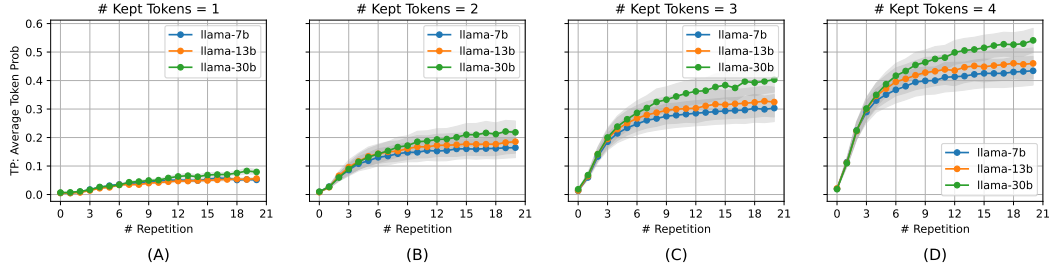


Figure 13: Token co-occurrence reinforcement on Wikitext-103.

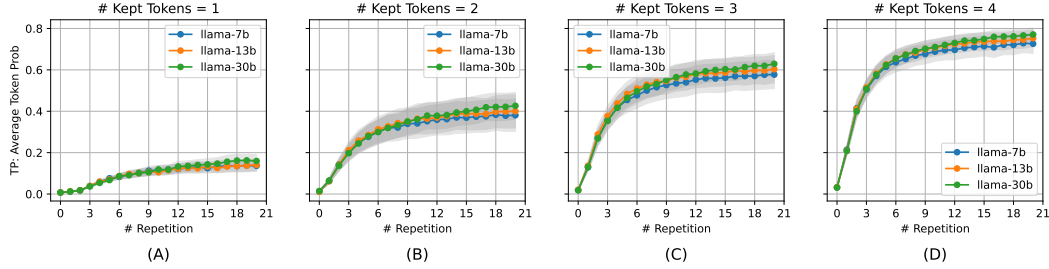


Figure 14: Token co-occurrence reinforcement on BookCorpus.

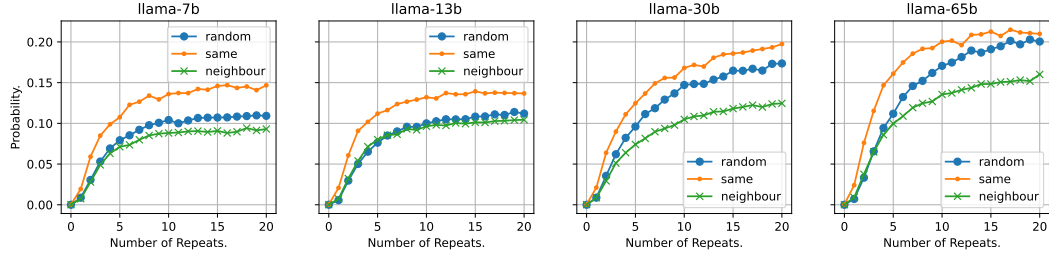


Figure 15: Token reinforcement against semantic relationships.

D.4 TOKEN REINFORCEMENT ON OTHER DATASETS

Here, we plot the token reinforcement on Wikitext-103 and BookCorpus. As we can see, the probability of reinforced tokens is quite different in the two datasets. In BookCorpus, with 4 kept tokens, the probability can be boosted to about 0.8, whereas in Wikitext-103, the probability can only reach about 0.5. Note that compared to the results on randomly generated sentences, tokens in these two datasets are more likely to co-occur in the pre-training data. This indicates the semantic relationship between tokens affects how token reinforcement performs.

D.5 SEMANTIC RELATIONSHIPS OF TOKEN REINFORCEMENT

We further conduct experiments to investigate how the semantic relationship between two tokens affects the token co-occurrence reinforcement. We choose three relationships: (1) Random two tokens. (2) The same two tokens. (3) Two tokens that are similar in the embedding space.

Figure 15 plots our results for different LLaMA models. We observe clear gaps among different semantic relationships. Two tokens that are the same can build a strong connection stronger than two tokens that are similar in the embedding space. Further investigating the reasons behind this is interesting and may unravel the internal biases of large language models. We leave it as the future work.

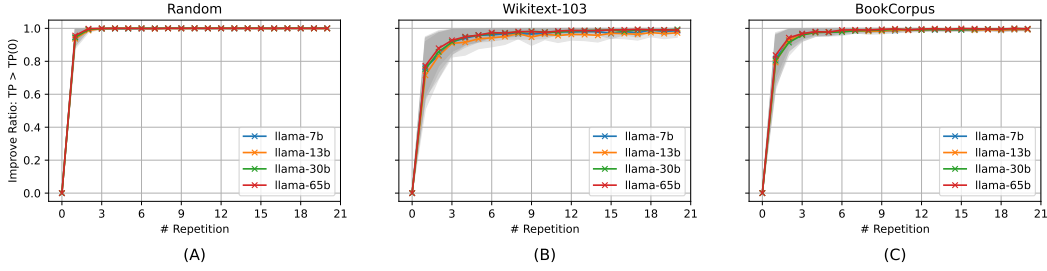
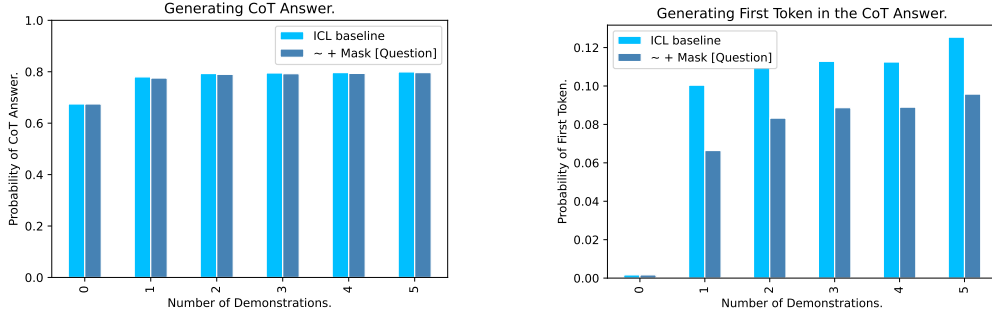


Figure 16: Improved Ratio on three datasets.

Figure 17: *Left: The probability of generating the CoT answer. Right: The probability of generating the first token in CoT answer.*

D.6 IMPROVED RATIO OF TOKEN REINFORCEMENT

In this section, we solidify our findings with the ratio of improved token probability. Formally, the improved ratio (IR; Xu et al. (2022)) is defined as follows:

$$\hat{\text{IR}} = \hat{\text{TP}}_N > \hat{\text{TP}}_0. \quad (2)$$

Our metric of improved ratio is defined over the setting in Section 3.1. Figure 16 plots our results of IR on three datasets. We only consider the case of two tokens.

As we can see, the improved ratios for all three datasets quickly reach almost 1.0 with only several repetitions, indicating token reinforcement exists in most of the cases. In addition, we observe that IRs in Wikitext-103 and BookCorpus have larger variances than those in randomly generated sentences, because tokens in Wikitext-103 and BookCorpus are more likely to co-occur in the same context and thus get larger probabilities without reinforcement.

E OTHER EXPERIMENTS ON DETRIMENTAL EFFECTS

E.1 GENERATION OF COT ANSWER

In this section, we study how the CoT answer is generated with respect to our discovered surface patterns. We study the problem on the GSM8K dataset. Recall the demonstrations in GSM8K, i.e., “Question: [Question] // Let’s think step by step. // [CoT Answer]”. We plot the probability of [CoT Answer] in Figure 17. We would expect the [Question] to have a huge effect on generating the CoT answer. Thus, we involve the probability with random substitution of the [Question] part. Interestingly, we find that even without the [Question] part, the probability of [CoT Answer] still increases with demonstrations, suggesting the patterns play a crucial role in learning how to generate the CoT answers.

So what does the [Question] part do in in-context learning of GSM8K? We further plot the probability of the first token of [CoT Answer]. The probability of the first token significantly decreases when we mask out [Question]. Hence, even though the [Question] part does not affect much the probability of [CoT Answer], it substantiates the generation of [CoT Answer] at the very beginning.